

## **Title of Project:**

# **ThreatShield DL: Deep Learning Based Web Attack Detection System**

## **Group**

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## **Abstract**

Cybersecurity threats have become increasingly complex as modern digital systems continue to expand and handle large volumes of network traffic and user requests. Detecting malicious activities within web-based environments is a critical challenge for organizations because attackers frequently exploit vulnerabilities such as SQL Injection, Cross-Site Scripting (XSS), and path traversal to gain unauthorized access, manipulate data, or disrupt services. Traditional security mechanisms mainly rely on predefined rules or signature-based detection methods, which often struggle to identify newly emerging attack patterns or variations of known threats. To address these limitations, this project proposes ThreatShield DL, a deep learning based cyber-attack detection system designed to analyze HTTP request patterns and identify suspicious behavior in real time. The proposed system utilizes a trained deep learning model capable of learning patterns from labeled datasets containing both benign and malicious request samples. By applying feature extraction and preprocessing techniques, the system converts incoming request data into a format suitable for machine learning analysis and classifies each request as either normal or potentially harmful. In addition to the deep learning model, rule-based verification mechanisms are incorporated to strengthen detection reliability and reduce the likelihood of false positives. The system is implemented through an interactive browser-based dashboard that allows users to submit HTTP requests for analysis and view the predicted attack category along with confidence scores generated by the model. These confidence scores help administrators evaluate the probability of malicious activity and support informed security decisions. Furthermore, the system provides basic explanations and recommended responses such as blocking suspicious inputs or monitoring traffic for further investigation. By integrating machine learning techniques with practical security monitoring tools, ThreatShield DL demonstrates an intelligent approach to identifying cyber threats and enhancing defensive capabilities. The proposed solution offers a lightweight, efficient, and deployable framework that can assist administrators in monitoring suspicious requests and improving the overall security posture of web-based systems.

**Keywords:** Cybersecurity, Deep Learning, Web Attack Detection, SQL Injection Detection, Cross-Site Scripting (XSS), HTTP Request Analysis, Security Monitoring.

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**Novelty:**

- Use of deep learning models to detect web application attacks instead of relying only on static rule-based security filters.
- Integration of AI-based attack classification with a web-based monitoring dashboard for real-time request analysis.
- Detection of multiple web attack categories such as SQL Injection, XSS, and path traversal using machine learning techniques.
- Provision of confidence scores for detected threats to help administrators evaluate the reliability of predictions.

**Add-ons:**

- Explainable alerts showing simple reasons for detected attacks (e.g., suspicious SQL patterns or script tags).
- Adaptive response mechanism suggesting actions such as blocking requests or monitoring suspicious traffic.
- Security logging system to record detected attacks for analysis and monitoring.
- Confidence-based threat evaluation to distinguish between high-risk and low-risk requests.
- Interactive web dashboard for analyzing HTTP requests and visualizing detection results.